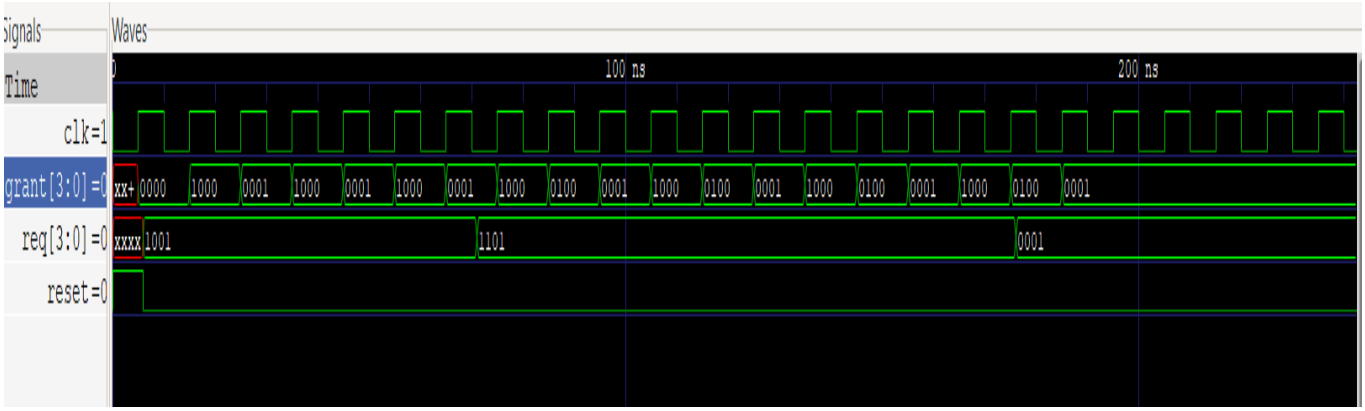
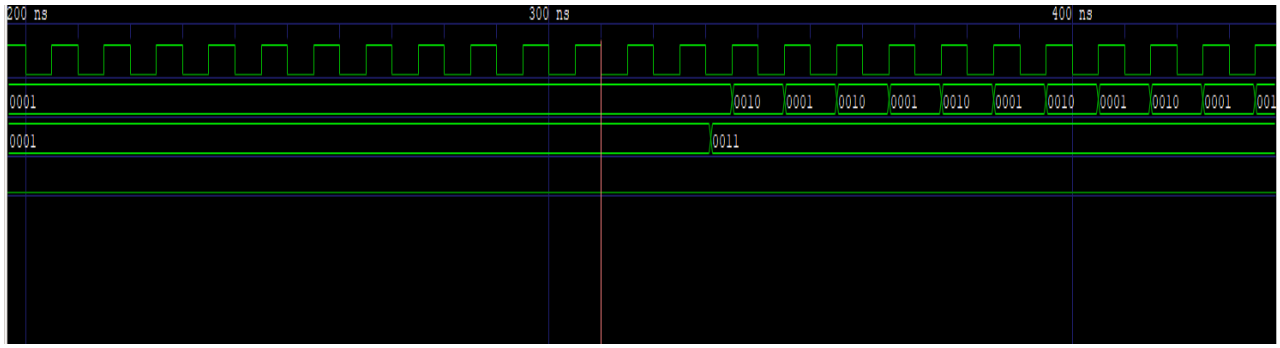
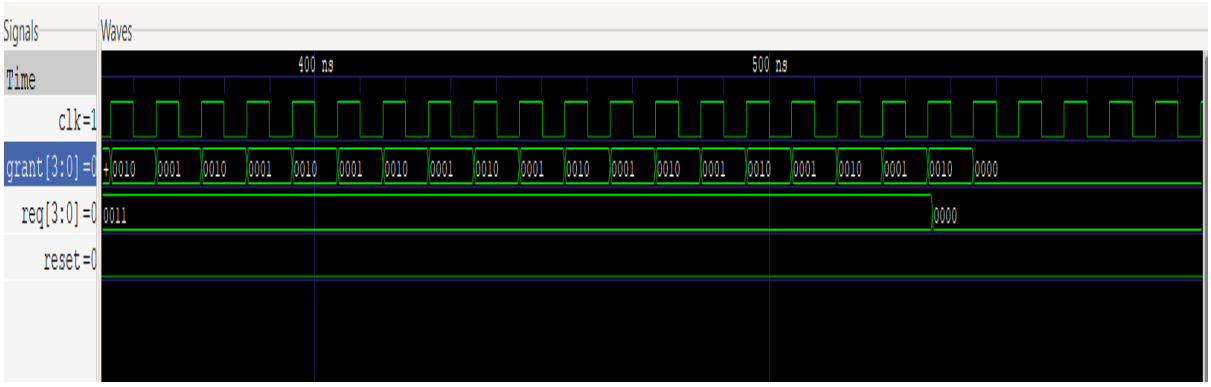






Explaination

Whole system works like a fsm where the grant changes only at clock transitions. 3,2,1,0 refers to req[3],req[2] ans so on.

the systems which are not requesting are completely ignored. At state no active grant the priority order is 3>2>1>0.

The grant priority changes in a cyclic manner and at success of grant 3 the highest priority then goes to 2 then 1 then 0 and at end 3.if at 2 then 1 and so on.at 0 the next priority goes to 3 as shown in images 1.

Grant can be hogged by one requester only if all the other 3 are not requesting.as seen at the end of 1st image.

Any req change does not show results utill a clock cycle end.the requests at the edge and current grant is what matters.

The vcd vvp and Verilog files are given along with this pdf